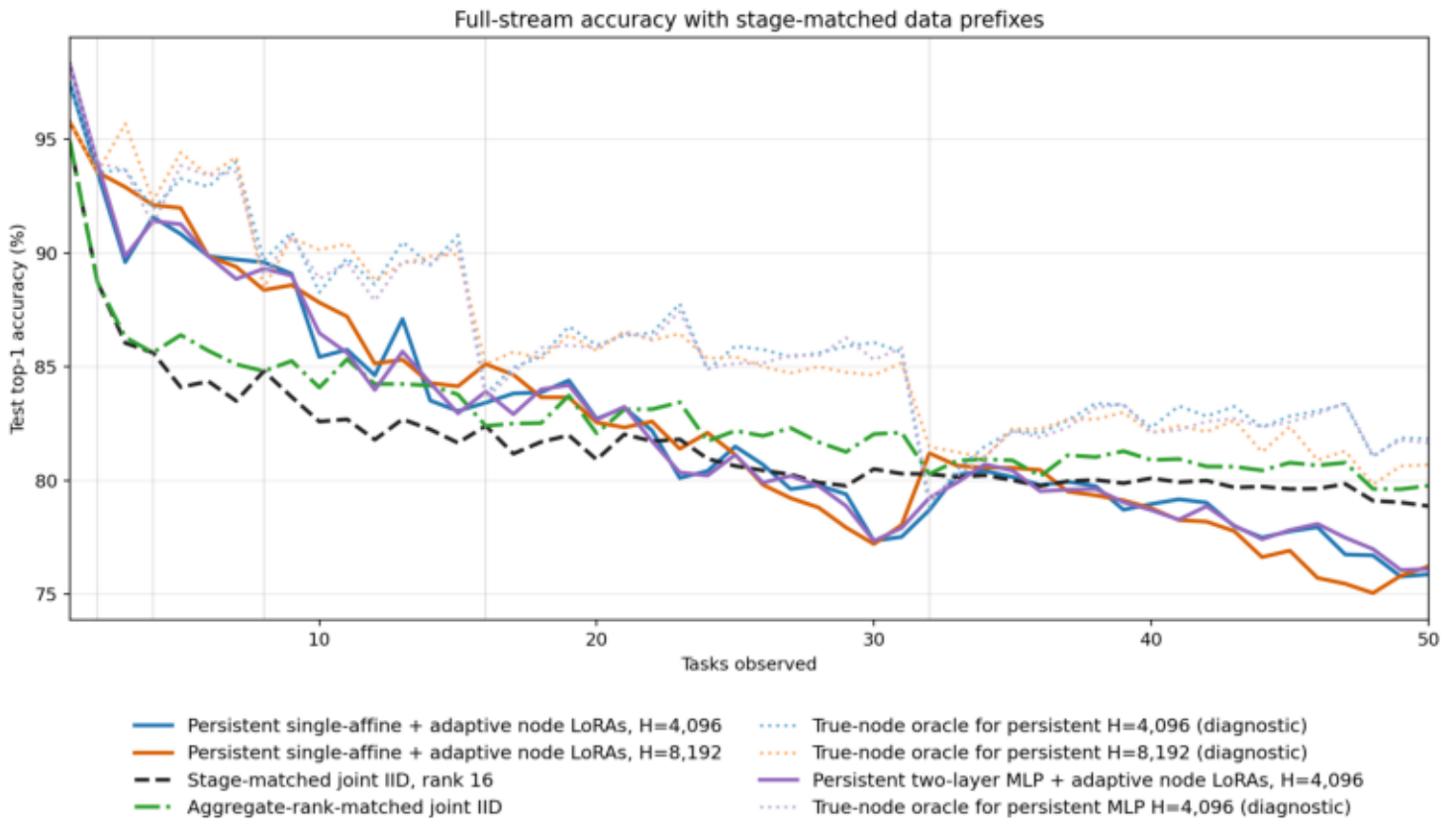


ImageNet-R-50 persistent frontier integrators

Single-affine H=4,096 / H=8,192 and two-layer MLP H=4,096

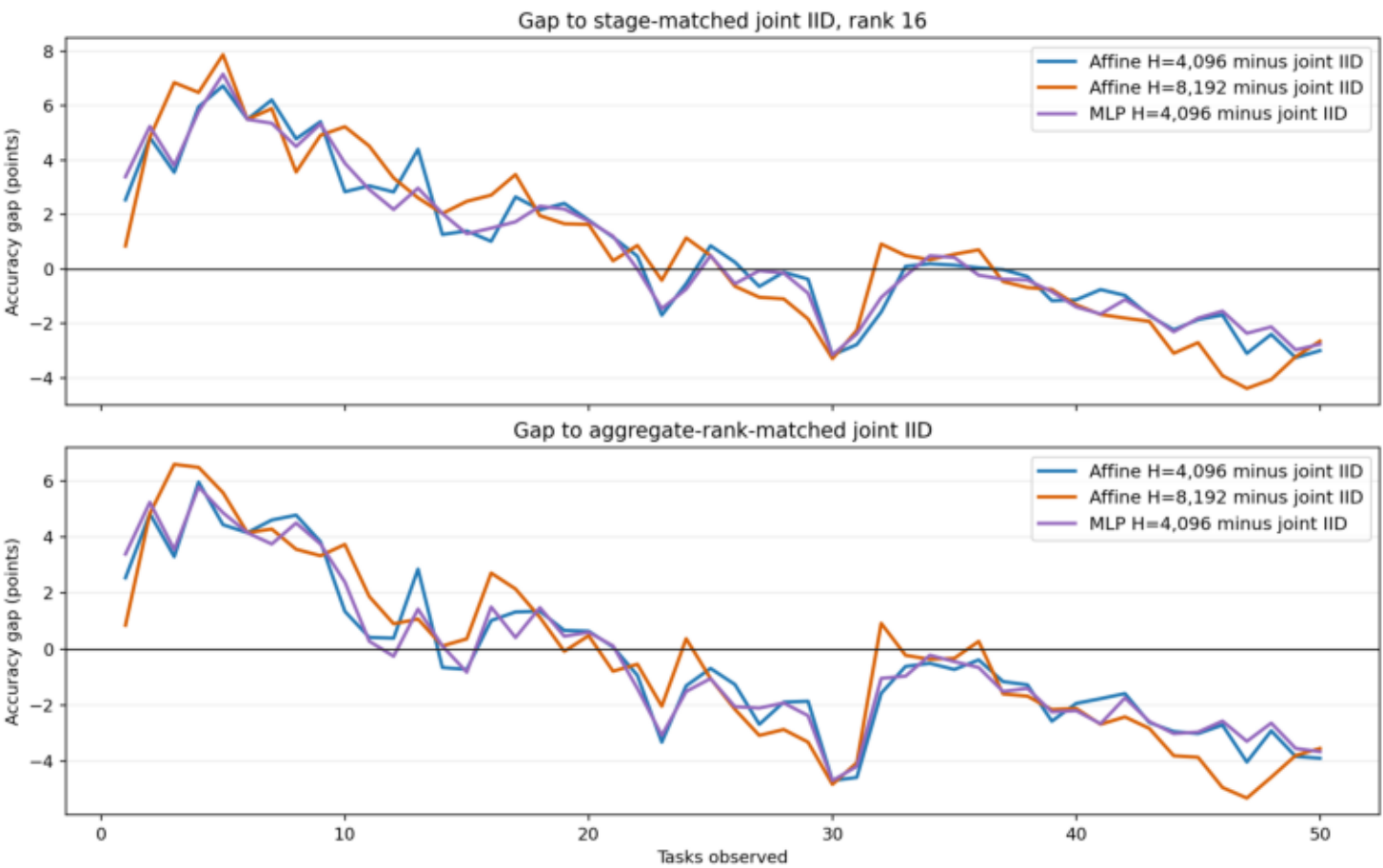
At task 50, MLP H=4,096 reaches 76.100% accuracy, versus 75.867% for affine H=4,096 and 76.217% for affine H=8,192. Rank-16 and aggregate-rank joint IID reach 78.867% and 79.767%. Mean accuracy over all 50 stages is 82.416% for MLP, 82.434% and 82.452% for the affine arms, and 81.630% and 82.759% for the joint curves.



Blue and orange are the single-affine H=4,096 and H=8,192 conditions. Purple is the two-layer ReLU MLP at H=4,096. All use each live node's own adapted 768-value pre-classifier latent, without macro tokens or metadata. Dotted matching colors are label-aware true-node diagnostics, not deployable conditions. Black is stage-matched rank-16 joint IID; green matches total live LoRA rank.

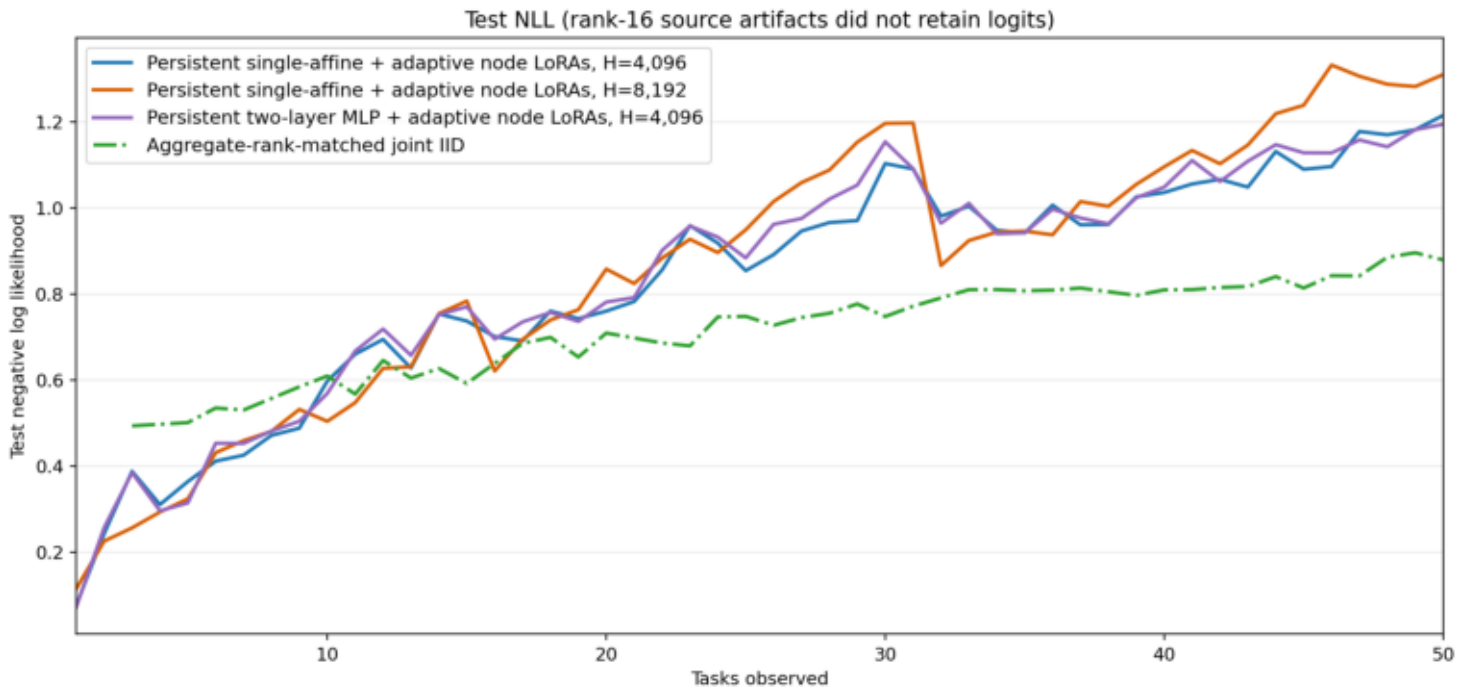
Selected stages and control gaps

Stage	Nodes	Affine 4k	Affine 8k	MLP 4k	MLP oracle	Joint r16	Rank-matched	Rank
1	1	97.46%	95.76%	98.31%	98.31%	94.92%	94.92%	16
2	1	93.55%	93.55%	93.95%	93.95%	88.71%	88.71%	16
4	1	91.58%	92.11%	91.40%	91.23%	85.61%	85.61%	16
8	1	89.59%	88.37%	89.31%	89.02%	84.80%	84.80%	16
16	1	83.41%	85.11%	83.90%	83.61%	82.40%	82.40%	16
31	5	77.52%	78.04%	77.91%	85.82%	80.29%	82.10%	80
32	1	78.69%	81.19%	79.23%	79.15%	80.28%	80.28%	16
50	3	75.87%	76.22%	76.10%	81.65%	78.87%	79.77%	48



Positive gap means the named integrator is ahead of the joint-IID reference. Power-of-two stages have one node. Table labels 4k/8k mean H=4,096/8,192; oracle is diagnostic only.

Likelihood and protocol



Rank-16 source artifacts retained hard predictions rather than logits, so rank-matched NLL is absent at stages 1, 2, 4, 8, 16, and 32. Accuracy remains available at all 50 stages.

Exact online update rule

Every arrival uses all current-task images plus a deterministic stage-keyed class-stratified historical draw. $H=4,096$ receives four passes and $H=8,192$ receives five. Dense-head parameters and AdamW state carry forward. A node adapter and its moments carry only if its hierarchy-node hash survives; new leaves and consolidation parents start from authenticated source adapters. The base ViT and local classifiers never train.

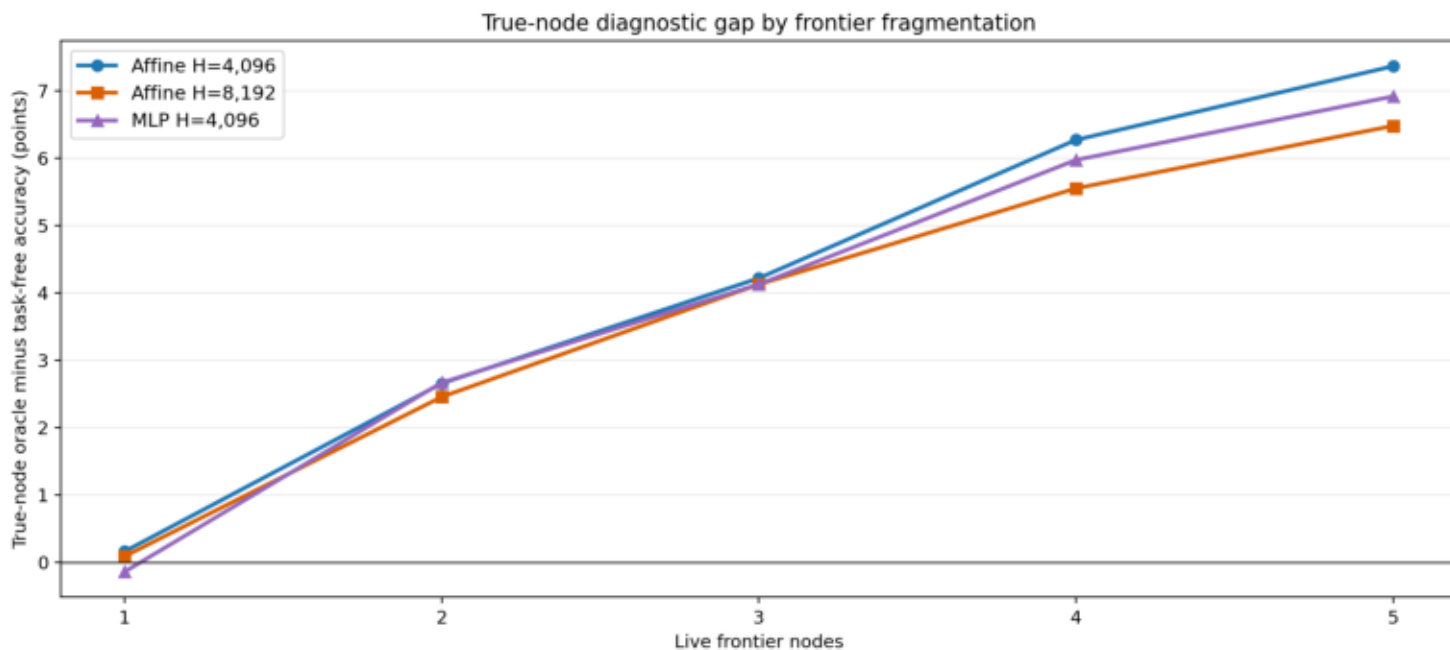
Comparison boundary

Stage-matched joint IID trains one fresh rank-16 adapter for five epochs on every prefix. Aggregate-rank-matched joint IID uses rank and alpha equal to 16 times popcount(stage). It matches total live LoRA rank, but not pretrained node state, multiple ViT paths, dense-head parameters, or deployment compute. Neither curve gates execution.

Original affine/joint provenance

Protocol 5f60e2f5c25e3318ba9c9b66d41ce8c159ba57a0f1a4fc524567c098b222d686
Result 4fa601c9b8e3cc87f07897f06ba51f88b75b7ecde1bf47ef01881526df4b4b58
Source hierarchy unchanged: True

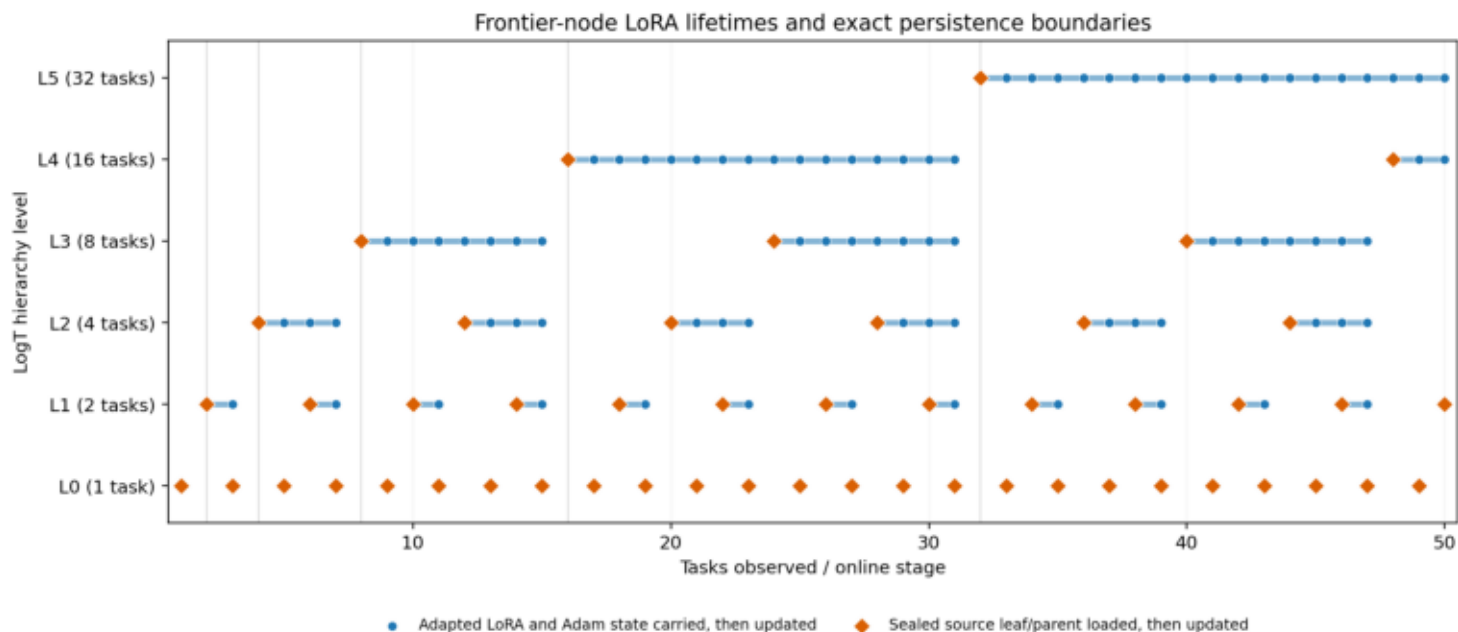
Fragmentation diagnosis



Nodes	Stages	Affine 4k acc.	Affine 4k gap	Affine 8k acc.	Affine 8k gap	MLP 4k acc.	MLP 4k gap
1	6	89.05%	0.16	89.35%	0.09	89.35%	-0.14
2	15	83.76%	2.66	84.31%	2.46	83.73%	2.67
3	18	81.39%	4.22	81.05%	4.13	81.15%	4.13
4	9	79.10%	6.27	78.82%	5.55	79.18%	5.98
5	2	77.13%	7.37	76.75%	6.48	77.70%	6.92

The label-aware true-node oracle is diagnostic only. Its growing advantage as the number of independently adapted frontier nodes rises is consistent with cross-node competition. Because it also substitutes the frozen node-local classifier, it does not isolate routing alone.

Frontier-node adaptation lifetimes



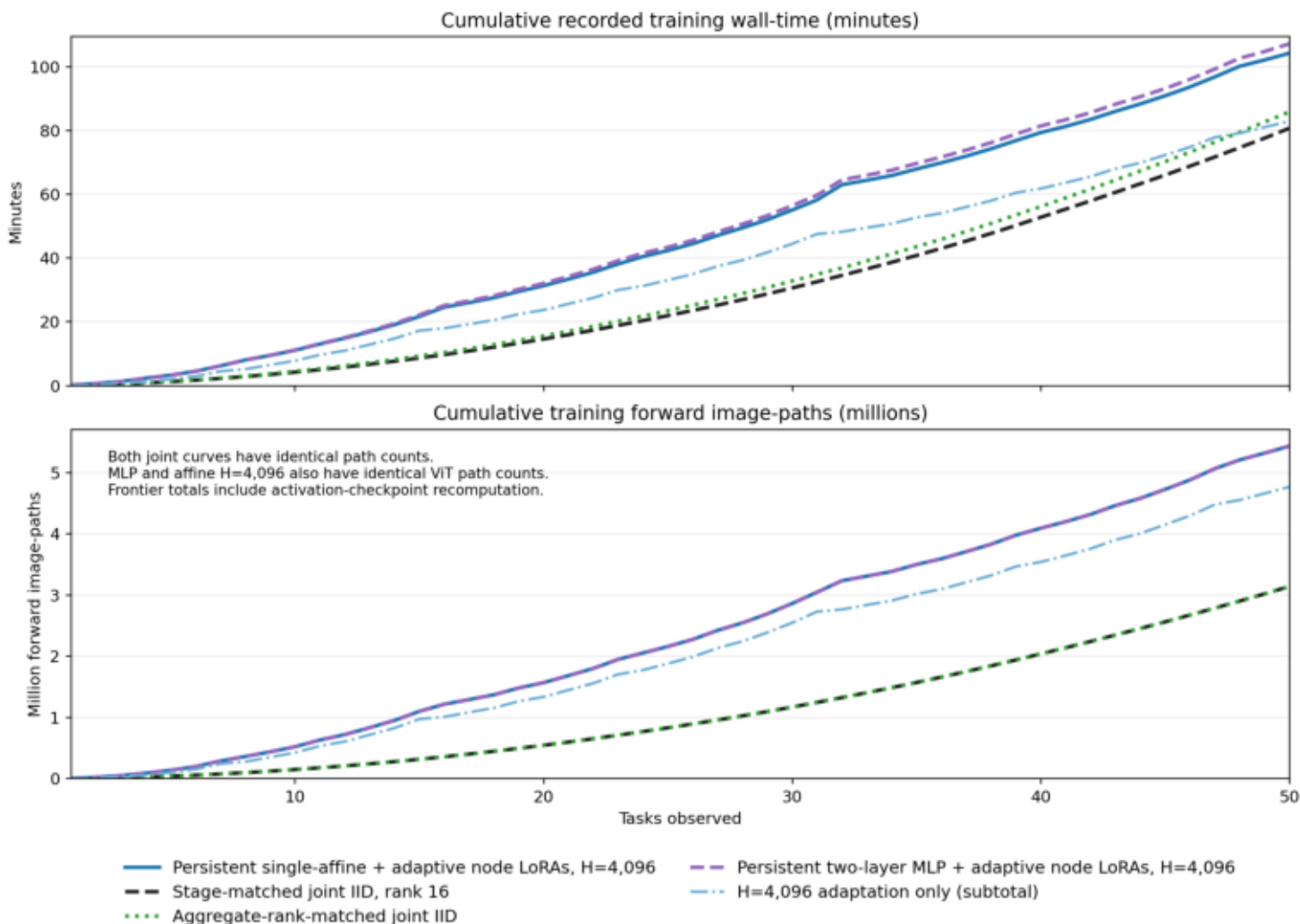
What persists

A blue lifetime continues only while the exact source hierarchy-node hash remains live. The next stage begins from that node's final adapted rank-16 LoRA, its 768-column head input block, and its named AdamW moments. All live LoRAs then update jointly on every current-plus-replay example. The affine and MLP conditions share these lifetimes but train their own independent parameters.

What resets

An orange diamond is a new leaf or consolidation parent loaded from the sealed source hierarchy. A parent was independently retrained on the full union of its represented tasks; it does not inherit the online-adapted LoRAs or optimizer moments of retired children. Unaffected nodes continue normally. Global head biases persist; in the MLP, shared hidden biases and old-class output rows also persist across consolidation.

Cumulative training cost



Training cost at 50 tasks

Affine H=4,096 costs 1.29 times the rank-16 joint curve's recorded training time and 1.73 times its forward image-path count including checkpoint recomputation. The hierarchy subtotal includes all 50 leaves and 47 parents, including intermediate parents created and retired in the same arrival. The smaller asymptotic bound has not produced a wall-time saving at this horizon. The MLP H=4,096 total is 107.19 recorded training minutes, with exactly the same ViT path counts as affine H=4,096.

What is measured

One image-path is one image processed by one ViT with one installed adapter. Forward and backward counts are reconstructed from completed image-presentation counters and actual live-node counts. Recomputation is one additional forward invocation per image/node under activation checkpointing; it can stop early. These are path counts, not profiled FLOPs or equal-cost forward/backward operations. The dense head (affine or MLP), optimizer, and differing LoRA ranks also affect wall time.

Resource accounting and scaling

Condition / component	Train min	Forward M	Recompute M	Backward M
Affine H=4,096 total	104.23	3.051	2.385	3.051
H4 hierarchy subtotal	21.33	0.665	0.000	0.665
H4 adaptation subtotal	82.90	2.385	2.385	2.385
Joint IID, rank 16	80.72	3.140	0.000	3.140
Joint IID, rank matched	85.88	3.140	0.000	3.140
MLP H=4,096 total	107.19	3.051	2.385	3.051

M = million image-paths. Recompute = activation checkpointing. Affine H4 has two subtotals.

Wall-time and reuse

Curves sum recorded training-job wall times, including data loading and checkpoint writes inside those timers. They exclude setup, artifact validation, inter-job overhead, and evaluation. Source training is charged when its subtree becomes available. The imported final rank-16 joint model is charged its original training cost; each reused reference is counted once within each condition. These are alternative algorithm totals, not a sum of actual work across the experiment invocations.

Why ViT training is $O(T \log T)$

Let $P_t = 4(m_t + \min(4096, M_{t-1}))$, where m_t is the new task size and M_t is the seen training prefix. With $k_t = \text{popcount}(t)$, adaptive forwards are $A_T = \sum(k_t P_t)$. Each hierarchy example belongs to at most one five-epoch job per level, giving $S_T = 5 \sum_v(n_v)$. Total forward work including checkpoint recomputation is $S_T + 2A_T$; backward work is $S_T + A_T$. For fixed replay capacity, epochs, model size, and bounded task size, both are $O(T \log T)$. The joint curves each use $5 \sum_t(M_t)$ forward/backward pairs, which is $O(T^2)$ in model paths.

The entire benchmarking workflow has a larger bound

Repeated full-prefix tests add 471,565 forward image-paths and 6.06 measured minutes for affine H=4,096. Each joint curve adds 157,149 test paths. Rank-16 test time is 2.55 minutes; rank-matched test wall-time was not retained. Testing every growing prefix is $O(T^2 \log T)$ for the frontier. The replay sampler also scans all earlier identities at each arrival, giving at least quadratic CPU bookkeeping. The $O(T \log T)$ claim therefore applies to ViT training, not the whole runner. Both head architectures have 200 outputs here; unbounded class growth would require separate head-cost accounting. MLP H=4,096 adds 6.22 minutes for the same test-path count.

Joint curves versus one final offline fit

Both joint curves refit at every prefix, so their base-model path counts coincide. The larger rank changes arithmetic per path, not the number of paths. A single final rank-16 offline fit cost 3.11 minutes and 120,000 forward/backward pairs; it provides only the final model and has linear training work in the total image count.

Two-layer MLP: matched H=4,096

Stage	Nodes	Affine acc.	MLP acc.	Affine NLL	MLP NLL	Affine oracle	MLP oracle
1	1	97.458%	98.305%	0.077	0.073	97.458%	98.305%
2	1	93.548%	93.952%	0.243	0.257	93.548%	93.952%
4	1	91.579%	91.404%	0.311	0.297	91.930%	91.228%
8	1	89.587%	89.306%	0.472	0.482	89.681%	89.024%
16	1	83.414%	83.899%	0.701	0.695	83.802%	83.608%
31	5	77.516%	77.909%	1.091	1.091	85.639%	85.823%
32	1	78.694%	79.229%	0.981	0.965	78.847%	79.153%
50	3	75.867%	76.100%	1.215	1.194	81.833%	81.650%

Measured effect of the second layer

At task 50 the MLP reaches 76.100% accuracy and 1.1945 NLL, a change of +0.233 accuracy points and -0.0206 NLL relative to affine H=4,096. Its mean accuracy change over the 44 fragmented stages is -0.061 points. Its final true-node diagnostic is 81.650%.

Architecture and size

Every input vector is the final 768-value latent from a live node's own base ViT plus adaptive rank-16 LoRA. The head is Linear(768 x live nodes, 1024), ReLU, then Linear(1024, 200). At five nodes it has 4,138,184 head parameters, versus 768,200 for affine. Both have 6,635,520 trainable LoRA parameters. There is no skip connection, normalization, dropout, metadata, score input, or macro token.

Initialization and continuing state

The first 400 hidden units split the 200 source-union logits into positive and negative parts; their signed output reproduces the source union. Another 624 units begin as random latent projections with zero output weights. Both dense matrices train freely. Surviving input blocks, hidden biases, old-class output rows, and AdamW moments persist. New node blocks and adapters initialize from the sealed source. The shared MLP output layer is not reset at consolidation.

What this comparison tests

The MLP uses exactly the affine H=4,096 replay identities, augmentation/order seeds, four epochs, batch 64, learning rates, and LoRA carry/reset boundaries. Both joint curves and both affine arms are reused unchanged. This tests the larger nonlinear head under the existing optimization budget, not whether a separately tuned MLP has converged. It is one seed, with no test-selected checkpoint. Rank-matched joint IID does not match the MLP's extra parameters or multiple ViT paths.

MLP protocol bb991cf93bd05d74a92d796735768c67e1f2e9b78e5b40e1c2ee314b29594fd6
MLP result e86a99f4ead8eb40be436a88b6dd927ef6732a64c02c78077dce32ffea2aaf8f